

Johannes Dreckhoff

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About me

I am a graduate student in theoretical and computational (bio)physics with a strong background in math, mathematical modeling, biophysics, numerical simulations and high-performance scientific computing. I am experienced in Python and JAX and eager to apply my analytical skills to a wide range of problems inside and outside of academia.

Skills & Tools

- Programming:** Strong background in Python and [JAX](#) as well as Numpy and SciPy, Git, GitHub, functional programming, JIT, numerical simulations, object-oriented programming, scientific computing
- ML Foundations:** Stochastic Differential Equations, linear algebra, optimisation algorithms, Monte Carlo Methods, stochastic modeling
- Other:** Rigorous mathematics, mathematical modeling, biophysics, statistics, translating models into software, data visualisation, data structures, lab work

Education & Relevant Courses

M.Sc. Physics, Heidelberg University (9/2024-12/2025, Grade 1.0)

- [Master thesis](#) in theoretical and computational biophysics, Schwarz-Group (Grade 1.0)

B.Sc. Physics, Heidelberg University (10/2019-7/2023, Grade 1.3)

- [Bachelor thesis](#) in theoretical biophysics, Mielke-Group (Grade 1.0)
- Internship in the Mielke-Group, on emergence of structure in non-equilibrium systems
- Exchange semester, Physics, University of Leiden (7/2023-2/2024)

Physics of Finance, University of Leiden, Dr. Diego Garlaschelli (2023-2024)

- Covered stochastic processes, market and pricing models, network theory and application of physical methods to financial systems

Abitur (A-level), Theodor Fliedner Gymnasium (8/2011-7/2019, Grade 1.0)

Projects, Research & Publications

Pattern Formation in Lipid Membranes ([Bachelor Thesis](#), 2023)

- Used statistical mechanics to constrain differential equation solutions for pattern formation in lipid membranes
- Developed mechanism explaining formation of sharp pattern

Structural Assembly and Curvature Generation in Clathrin Networks ([Master Thesis](#), 2025)

- Developed python based differentiable framework for the kinetic simulation of clathrin assembly and curvature generation in cellular endocytosis
- Using JAX for high performance (JIT), cluster-compatibility, scalability and differentiability
- To be published on [GitHub](#) (Code samples available upon request)

Work Experience & Internships

Scientific Research Assistant, Heidelberg University (12/2025 - Present)

- Conducting research building upon my master thesis, refining the JAX framework, creating documentation, preparing manuscript for publication

Laboratory Assistant, Heidelberg University (2025)

- Guiding undergraduates of medicine/life sciences through laboratory experiments, laboratory documentation and evaluation

Tutor for Physics, Heidelberg University (2024)

- Leading tutorials for students of medicine/life sciences, communicating and explaining content of lectures and assisting students in solving and understanding problem sheets

Internship @ German Aerospace Center (DLR) (2018)

Internship @ Engineering Office „Baues & Wicht“ (2018)

Invited Talks

@ Institute for Science and Technology Austria (ISTA):

- How Clathrin Bends Membranes: Insights into Structural Coat Assembly from Agent-Based Modeling

Prizes & Awards

German Physical Society (DPG) Abitur & Book Prize (2018)

- For excellent and outstanding achievements in physics

German Mathematical Society (DMV) Abitur Prize (2018)

- For outstanding achievements in mathematics

Third place @ changes.AWARD (2018)

- For creating an innovative start-up business plan with focus on sustainability

Languages

German: Native

English: Fluent (C1-C2)

French: Intermediate (B1)

Interests

Team sports, Ultimate frisbee, Music (Guitar & Piano)